Learning to Find Cups: Reinforcement Learning for a Physical Ping-Pong Ball Launcher*

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Abstract

We present **Pong-Bot**, a physical reinforcement learning system in which a Raspberry Pi-controlled 3-axis CNC-style launcher fires ping-pong balls across a table to locate cups placed on a 6×5 grid. The robot receives only directional feedback (*left*/*right*/ *short*/*long*/*hit*) and has no camera or distance magnitude signal, making this a partially observable sequential decision problem with a high physical cost per sample. We propose a recurrent policy based on a Gated Recurrent Unit (GRU) trained via behavioural cloning (BC) from a simple heuristic baseline, followed by REINFORCE fine-tuning. The GRU integrates direction history across shots within an episode, enabling more efficient convergence than the stateless heuristic. In live evaluation over 30 sessions, the deterministic GRU policy achieves 100% hit rate with a mean of 7.7 shots, compared to 9.3 shots for the heuristic (both 100% hit rate). We further extend the framework with an outer GRU layer that provides warm start positions for multi-cup sessions, reporting preliminary results from 16 live three-cup sessions.

1 Introduction

Physical robot learning presents a fundamental challenge: environment interactions are expensive, noisy, and slow. A robot that must fire a projectile to locate an unknown target exemplifies this challenge. Each trial requires mechanical actuation, and the feedback signal is coarse. We study a concrete instantiation of this problem: a 3-axis CNC-style launcher controlled by a Raspberry Pi must find the location of a cup placed somewhere on a 6×5 grid of positions on a six-foot folding table. The only feedback available after each shot is a directional hint, *left*, *right*, *short*, *long*, or *hit*, with no distance magnitude and no camera. The cup position is unknown at session start, and the objective is to find it in as few shots as possible.

A simple heuristic policy provides an adequate starting point: it applies fixed directional corrections and achieves 100% hit rate. However, it cannot adapt its step sizes based on accumulated shot history, leaving room for a more efficient learned policy. Training a policy purely from reward signals without demonstrations would require far more physical robot interactions than are practical, since each interaction requires mechanical actuation and physical setup time. A static supervised model also falls short: with no memory of prior shots it cannot integrate directional evidence across an episode.

*. Code available at: https://github.com/rschmi3/pong-bot

Our approach combines three components. First, the heuristic provides 30 reliable demonstrations that solve the cold-start problem at zero training cost. Second, we train a GRU policy by behavioural cloning (BC) from these demonstrations; the GRU's recurrent hidden state integrates directional history across shots within an episode, enabling adaptive behaviour the stateless heuristic cannot perform. Third, we fine-tune with REINFORCE using collected GRU episodes, further reducing the number of shots to hit with only a small amount of additional robot data.

We additionally extend the system with an outer RL layer for multi-cup sessions. When multiple cups are placed on the table, the outer GRU selects a starting motor position for the next inner search based on the previous cup's winning position, providing a warm start in a region of the table not yet searched.

2 Techniques to Tackle the Problem

2.1 Related Work

Sequential search and Bayesian optimisation. Bayesian optimisation selects queries to balance exploration and exploitation over an unknown function; Srinivas et al. (2010) propose GP-UCB, a principled variant that builds a Gaussian process model of the objective. These methods require a scalar reward at each query. Our setting provides only a discrete directional hint per shot, left, right, short, or long, rather than a scalar value, making standard Bayesian optimisation difficult to apply directly.

Imitation learning and behavioural cloning. BC trains a policy to directly imitate expert demonstrations (Pomerleau, 1988). More sophisticated approaches such as DAgger (Ross et al., 2011) address distribution shift by iteratively querying the expert on states visited by the learner, but require repeated expert access during training. Since our heuristic demonstrations cover the search space well and distribution shift is mild, simple BC is sufficient and more practical given limited robot time.

Recurrent policies for POMDPs. In a Partially Observable MDP (POMDP), the agent cannot directly observe the true environment state and must infer it from a sequence of partial observations. Our problem is a POMDP: the cup position is never directly observed, and each shot reveals only a directional hint. Hausknecht and Stone (2015) demonstrate that replacing a memoryless policy with an LSTM allows an agent to solve partially observable Atari games by integrating observation history. Wierstra et al. (2010) show recurrent policy gradients outperform memoryless policies on POMDPs more generally. We use a GRU rather than an LSTM because our episodes are short (typically 3-16 shots), and the GRU's simpler single-state architecture trains faster with fewer parameters while achieving comparable performance on short sequences (Chung et al., 2014).

Policy gradient and REINFORCE. REINFORCE (Williams, 1992) is a policy gradient algorithm that updates the policy by reinforcing actions that led to high returns. We apply it for fine-tuning after BC pre-training, using conservative settings to avoid overfitting on our small number of collected episodes.

Physical robot RL with limited data. PILCO (Deisenroth and Rasmussen, 2011) is a model-based RL method that learns a model of the environment's dynamics to reduce

the number of physical interactions needed. While appealing for physical robots, it requires the environment to be modelable. In our setting, the mapping from motor position to directional feedback depends on ballistic physics and mechanical noise, making it difficult to model reliably. BC from heuristic demonstrations is a more practical alternative.

Hierarchical RL. Kulkarni et al. (2016) introduce hierarchical deep RL with sub-goal policies, where a high-level policy sets goals for a low-level policy to execute. Our outer/inner structure is analogous: the outer GRU provides a warm start position (high-level) while the inner GRU executes the search from that position (low-level). Unlike option frameworks, our inner policy is fixed during outer training, simplifying the learning problem.

2.2 Method

Problem as MDP. We formulate cup finding as a sequential decision problem with a fixed maximum number of steps. The state at step t is an 8-dimensional vector: normalised X and Y motor positions (each divided by 50,000 steps), a 4-dimensional one-hot direction encoding $[\text{left}, \text{right}, \text{short}, \text{long}]$ (all zeros at episode start before the first shot), a binary flag indicating whether the current shot scored a hit, and a normalised shot counter. Actions are continuous delta steps $(\Delta x, \Delta y)$ added to the current motor position. Episodes terminate on a hit or a configurable maximum shot count.

GRU policy architecture. The policy is a single-layer GRU with input dimension 8 (matching the state vector above), hidden dimension 64, and a linear output head producing 2-dimensional normalised deltas. The hidden state persists across shots within an episode, allowing the policy to maintain a running belief about the cup's location. At evaluation time the policy outputs a deterministic action; during data collection, small Gaussian noise ($\sigma \in \{0.005, 0.01\}$) is added to the output to encourage exploration. In both cases the output is scaled back to motor steps before execution.

Behavioural cloning training. We collect 30 heuristic sessions (all hits, 2-16 shots each) and train the GRU to match the heuristic's actions using L1 loss. L1 is preferred over MSE because early-episode corrections are large, and MSE's quadratic penalty would push predictions toward a conservative average. Two augmentations improve generalisation: (1) random horizontal flip (p=0.5), which mirrors the table geometry and doubles the effective dataset size; (2) an auxiliary loss that penalises actions moving in the wrong direction relative to the hint. Training uses Adam with adaptive learning rate scheduling for approximately 200 epochs.

REINFORCE fine-tuning. After BC pre-training, we apply one round of REINFORCE using all collected GRU sessions as training data. Each shot receives a reward of $+1$ on a hit and -0.05 otherwise; returns are discounted with γ=0.95 and normalised within each episode. We use 5 epochs at learning rate 10^{-5}. A second REINFORCE round consistently degraded performance, a sign of overfitting on the small number of episodes, so the final pipeline is BC followed by a single REINFORCE round.

Offline evaluation. To validate policies without physical robot access, we implement an offline simulator that replays known cup positions against a policy. Directional feedback is

computed geometrically by comparing each shot's motor position to the stored cup position, eliminating mechanical noise. This allows rapid policy evaluation and hyperparameter selection without robot time.

Outer RL extension. For multi-cup sessions, an outer GRU provides a warm start position for each inner search based on the previous cup's winning motor position. The outer policy is initialised via BC on 500 synthetic episodes, each consisting of 3 distinct cup positions drawn from the heuristic inner session records, then fine-tuned on live sessions where both the inner and outer policies are GRU-based.

2.3 Heuristic Baseline

The heuristic policy implements a coarse-to-fine binary-search strategy. For each direction hint it applies a fixed correction: left$\rightarrow+\Delta_x$, right$\rightarrow-\Delta_x$, short$\rightarrow+\Delta_y$, long$\rightarrow-\Delta_y$, with step sizes decreasing over the course of the episode. The X and Y axes are corrected independently. The policy is fully deterministic, requires no training data, and serves as both the performance baseline and the data source for BC.

3 Evaluation

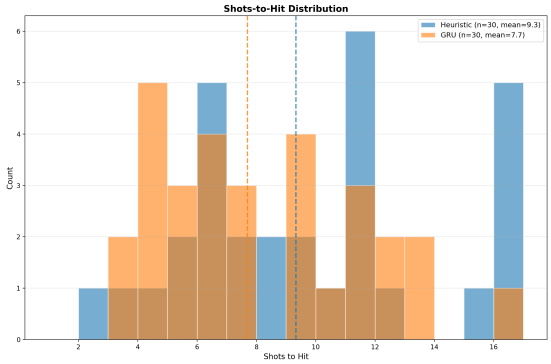
3.1 Inner Policy: Live Sessions

We evaluate three policy configurations on live sessions targeting single cups drawn from the 6×5 grid. Table 1 summarises results.

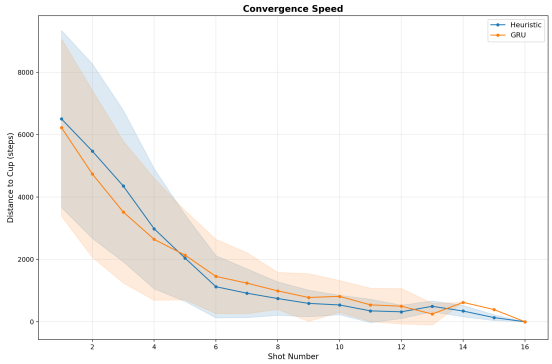
Table 1: Inner policy live session results.

Policy	N	Hit%	Mean	Med	Std	Min	Max
Heuristic	30	100%	9.3	9.0	4.2	2	16
GRU (σ>0)	36	83%	10.1	9.0	4.1	3	19
GRU (σ=0)	30	100%	7.7	7.0	3.5	3	16

The GRU policy at σ=0 matches the heuristic's 100% hit rate while reducing mean shots from 9.3 to 7.7, a 17% reduction. The exploratory GRU (σ>0) shows an 83% hit rate and higher mean shots, consistent with exploration noise degrading precision on some sessions.



(a) Shots-to-hit distribution for heuristic (blue) and GRU σ=0 (orange). Dashed lines indicate respective means. The GRU distribution is shifted left, with more mass at 4–7 shots.



(b) Mean distance-to-cup (motor steps) vs. shot number with ±1 std bands. The GRU converges faster in shots 2–5, then tracks the heuristic in the fine-convergence phase.

Figure 1: Inner policy performance.

3.2 Offline Simulation

Table 2 reports offline simulation results on the 30 heuristic cup positions. The simulator computes directional feedback analytically, eliminating mechanical noise.

Table 2: Offline simulation results (30 cup positions).

Policy	Hit Rate	Mean Shots
Heuristic	100% (30/30)	5.5
GRU σ=0	100% (30/30)	6.4

In the noiseless simulator the heuristic achieves 5.5 mean shots versus 6.4 for the GRU. This reversal relative to live sessions suggests that the heuristic's fixed step-size strategy is well-suited to noiseless conditions, while the GRU's learned behaviour may provide an advantage on the physical robot where mechanical variability is present. Directional accuracy in simulation is 100% across all four directions (left 14/14, right 56/56, short 7/7, long 85/85).

3.3 Inner GRU: Trajectory and Coverage

Figure 2 illustrates how the two policies navigate toward a cup across multiple sessions. The heuristic follows a systematic correction pattern while the GRU tends to converge more directly, consistent with its lower mean shots. Both policies achieve 0 timeouts across all 30 sessions, confirming reliable coverage of the full 6×5 grid.

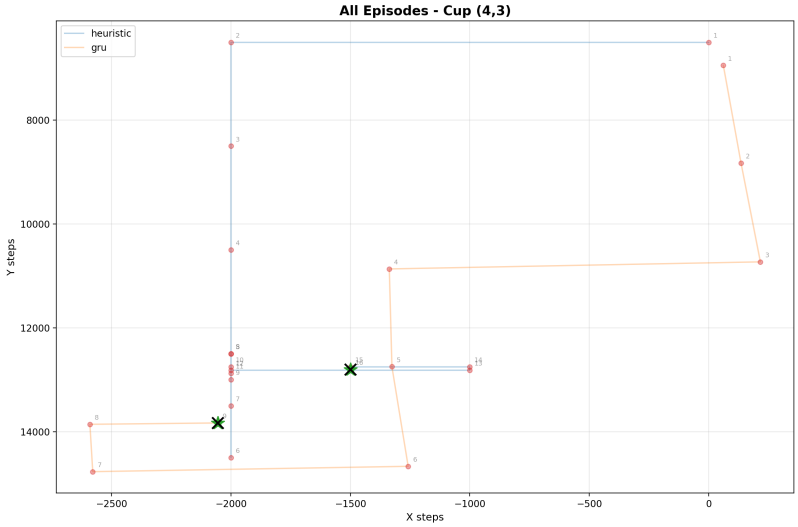


Figure 2: Shot trajectories for all sessions targeting cup (4,3). Blue: heuristic; orange: GRU. The GRU takes more direct paths.

3.4 Outer RL: Multi-Cup Sequencing

We evaluate the outer GRU on 3-cup sessions where the inner policy is also GRU-based, comparing against a simulated heuristic baseline. Table 3 and Figure 3 report results. No live heuristic outer sessions were collected; the baseline is a simulated distribution of 500 episodes, each consisting of 3 distinct cup positions drawn from the heuristic inner session records.

Table 3: Outer RL session results (3-cup sequences).

Configuration	N	Mean	Std	Min	Max
Heuristic + fixed outer (simulated)	500	28.1	6.9	10	48
GRU + GRU outer (live)	16	31.2	11.2	12	57

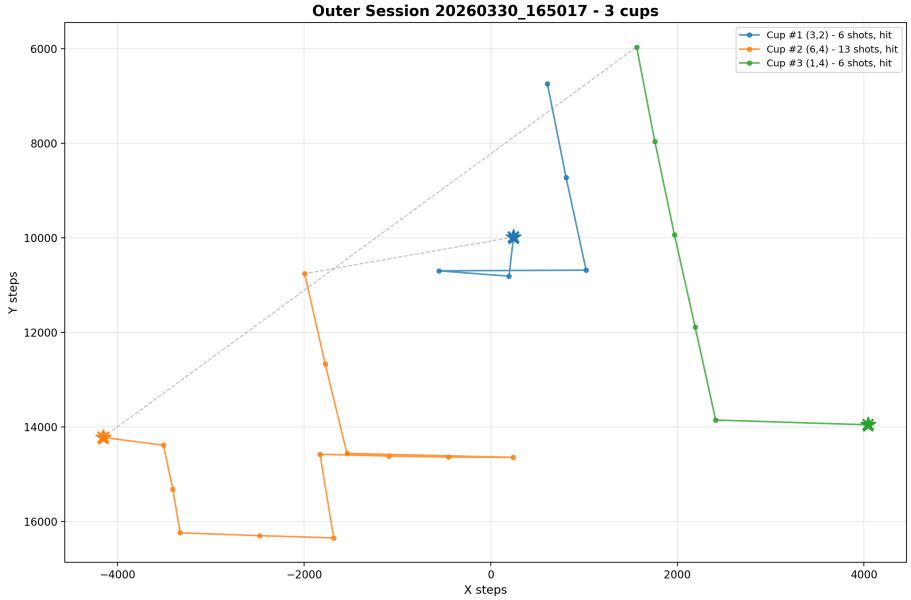


Figure 3: Example 3-cup outer session. Dashed lines show the warm start position provided by the outer GRU for each subsequent inner search; solid lines show the inner GRU's search trajectory for each cup.

The live configuration with GRU inner and outer policies (mean 31.2 shots, n=16) does not yet beat the simulated heuristic baseline (mean 28.1 shots, n=500). This is an early-stage result: the outer GRU has received only 16 live sessions and no REINFORCE fine-tuning. The comparison is further complicated by different noise profiles, live sessions include mechanical variability absent from simulation. We expect the gap to close with more data and REINFORCE fine-tuning.

4 Conclusion

We have presented Pong-Bot, a physical RL system that learns to find cups on a table by firing ping-pong balls and receiving only directional feedback. The GRU policy, trained via BC from heuristic demonstrations and fine-tuned with REINFORCE, achieves 100% hit rate with 7.7 mean shots on live sessions, compared to 9.3 mean shots for the heuristic baseline, a 17% improvement while maintaining perfect reliability. The BC+REINFORCE pipeline requires only 30 physical robot sessions, showing that a reliable heuristic can substitute for large amounts of robot time.

The outer GRU extension shows that warm-start sequencing across multiple cups is feasible. With 16 live sessions, the outer GRU approaches but does not yet surpass the simulated baseline, indicating the approach is promising but requires more data.

Limitations. The outer RL comparison is limited by different noise profiles between live and simulated sessions, making direct performance comparison difficult. The inner policy evaluation relies on 30 sessions, providing limited statistical power for fine-grained compar-

isons. Results may not generalise beyond the specific motor kinematics and table geometry used in this work.

Future work. Four directions are most promising. First, collecting more outer GRU live sessions and applying REINFORCE fine-tuning should close the gap with the simulated baseline. Second, transformer or other sequence model architectures could replace the GRU for more expressive history integration. Third, online adaptation, updating the policy continuously as new sessions are collected, could reduce the need for periodic offline retraining. Fourth, curriculum learning, starting with easier central cups and progressively introducing harder edge positions, may improve both sample efficiency and worst-case performance.

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